



GenomeInsight

Comprehensive Genetic Analysis Report

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File: MyHeritage_raw_dna_data.csv

Variants: 584,894 total, 423 analyzed

Report Overview

File: MyHeritage_raw_dna_data.csv • 584,894 variants in file • 423 analyzed



243

Health Risks

 111 actionable



31

Pharma

 12 actionable



3

Carrier



239

Traits

Health Risks

243 variants analyzed • 111 elevated

Cardiovascular

GENE	CONDITION	GENOTYPE	RISK LEVEL
CDKN2A/B	Coronary Artery Disease	GG	 Moderately Increased
9p21.3	Coronary Artery Disease	CG	 Moderately Increased
MTHFR	Elevated Homocysteine	AG	 Average
F2	Venous Thromboembolism	GG	 Average
F5	Factor V Leiden Thrombophilia	CC	 Average
4q25	Atrial Fibrillation	CC	 Average
4q25	Atrial Fibrillation	GG	 Average
LPA	Elevated Lipoprotein(a)	TT	 Average
AGT	Hypertension	AG	 Average
ADD1	Hypertension (salt-sensitive)	GG	 Average
AGTR1	Hypertension	AA	 Average
ADRB1	Heart Failure / Hypertension	CG	 Slightly Increased
HNF1A	Coronary Artery Disease	GG	 Average
WDR12	Coronary Artery Disease	TT	 Average
LDLR	Elevated LDL Cholesterol	CC	 Average
LDLR	Elevated LDL Cholesterol	GG	 Low
APOB	Elevated LDL Cholesterol	CC	 Slightly Increased
APOA5	Elevated Triglycerides	AA	 Average
NINJ2	Stroke (Ischemic)	AG	 Slightly Increased
HSPB7	Heart Failure	TT	 Average
TRIB1	Elevated Triglycerides	AA	 Slightly Increased
PCSK9	Coronary Artery Disease (protective)	GG	 Average
SORT1/CELSR2	Elevated LDL Cholesterol / CAD	TC	 Slightly Increased
PCSK9	Elevated LDL Cholesterol	TT	 Slightly Increased
ABO	Venous Thromboembolism	CC	 Average
CYP17A1	Hypertension	TT	 Slightly Increased
HMGCR	Elevated LDL Cholesterol	CC	 Average
LPL	Triglycerides / CAD (protective)	CC	 Average
CETP	HDL Cholesterol	AG	 Slightly Increased
APOC1	Coronary Artery Disease / Lipids	AA	 Average
NOS1AP	Long QT Syndrome	TC	 Average
LDLR	Elevated LDL Cholesterol	TT	 Average
MRAS	Coronary Artery Disease	CC	 Average
PPAP2B	Coronary Artery Disease	AA	 Moderately Increased

GENE	CONDITION	GENOTYPE	RISK LEVEL
MIA3	Coronary Artery Disease	AA	 Average
ZC3HC1	Coronary Artery Disease	TC	 Slightly Increased
HDAC9	Ischemic Stroke (Large Vessel)	GA	 Moderately Increased
PHACTR1	Coronary Artery Disease	GG	 Average
CXCL12	Coronary Artery Disease	CT	 Slightly Increased
SORT1/PSRC1	Elevated LDL Cholesterol	AG	 Slightly Increased
ATP2B1	Hypertension	GA	 Slightly Increased
NT5C2	Hypertension	TT	 Slightly Increased
NPR3	Hypertension	AA	 Average
FGF5	Hypertension	GG	 Average
CYP17A1 - CNNM2	Coronary Artery Disease	GG	 Slightly Increased
SORT1	LDL Cholesterol / Myocardial Infarction	GT	 Slightly Increased
APOA5 - ZNF259	Elevated Triglycerides	CC	 Average
CETP	HDL Cholesterol	AC	 Slightly Increased
9p21.3	Coronary Artery Disease / Aneurysm	GA	 Slightly Increased
GALNT2	HDL Cholesterol	GA	 Slightly Increased

Metabolic

GENE	CONDITION	GENOTYPE	RISK LEVEL
IGF2BP2	Type 2 Diabetes	GT	 Slightly Increased
TCF7L2	Type 2 Diabetes	TC	 Moderately Increased
PPARG	Type 2 Diabetes	CC	 Slightly Increased
KCNJ11	Type 2 Diabetes	TT	 Moderately Increased
SLC30A8	Type 2 Diabetes	CC	 Slightly Increased
CDKAL1	Type 2 Diabetes	GG	 Average
FTO	Obesity	TT	 Average
MC4R	Obesity	CC	 Slightly Increased
FTO	Obesity	TT	 Average
TMEM18	Obesity	CC	 Slightly Increased
GCKR	Elevated Triglycerides	CC	 Average
PNPLA3	Non-Alcoholic Fatty Liver Disease	CC	 Average
DHCR7/NADSYN1	Vitamin D Deficiency	GT	 Slightly Increased
MCM6/LCT	Lactose Intolerance	GG	 Average
GCKR	Gout	CC	 Average
ABCG2	Gout	GG	 Average
HFE	Iron Overload (H63D)	CC	 Average
TMPRSS6	Iron Deficiency Anemia	AA	 Average
TM6SF2	NAFLD / Liver Fibrosis	CC	 Average
HHEX	Type 2 Diabetes	TC	 Slightly Increased

GENE	CONDITION	GENOTYPE	RISK LEVEL
CDKAL1	Type 2 Diabetes	AA	 Average
GNPDA2	Obesity	AA	 Average
BDNF	Obesity / Hyperphagia	CC	 Average
MTNR1B	Type 2 Diabetes / Fasting Glucose	GC	 Slightly Increased
GCK	Fasting Glucose	GG	 Average
THADA	Type 2 Diabetes	TT	 Moderately Increased
ADIPOQ	Type 2 Diabetes / Metabolic Syndrome	GG	 Average
MTRR	Elevated Homocysteine / Neural Tube Defects	AA	 Average
MTNR1B	Elevated Fasting Glucose	TC	 Slightly Increased
KLF14	Type 2 Diabetes	AG	 Slightly Increased
TP53INP1	Type 2 Diabetes	TC	 Slightly Increased
ADRB3	Metabolic Syndrome / Obesity	GA	 Average
ENPP1	Type 2 Diabetes / Insulin Resistance	AA	 Average
PROX1	Type 2 Diabetes	TC	 Slightly Increased
ADRA2A	Elevated Fasting Glucose	GG	 Slightly Increased
FADS1	Dyslipidemia	TC	 Slightly Increased
PPARA	Dyslipidemia / Metabolic Syndrome	CC	 Slightly Increased
IRS1	Type 2 Diabetes / Insulin Resistance	CC	 Moderately Increased
GCKR	Fasting Glucose / Triglycerides	CC	 Average
ADAMTS9	Type 2 Diabetes	TT	 Average
PSMD6	Hypertension	TT	 Average
MTNR1B	Gestational Diabetes	GC	 Slightly Increased
IRS1	Insulin Resistance / CAD	AA	 Slightly Increased
KCNQ1	Type 2 Diabetes	CC	 Moderately Increased
FTO	Obesity	CC	 Average

Cancer

GENE	CONDITION	GENOTYPE	RISK LEVEL
BRCA1	Breast Cancer Risk	TT	 Average
CYP1A1	Lung Cancer Susceptibility	TT	 Average
8q24	Colorectal Cancer	GT	 Slightly Increased
8q24	Prostate Cancer	CC	 Average
TERT-CLPTM1L	Bladder Cancer	CT	 Slightly Increased
TERT	Lung Adenocarcinoma	AC	 Slightly Increased
MYNN/TERC	Colorectal Cancer	CC	 Slightly Increased
8q24	Breast Cancer	GG	 Slightly Increased
TOX3	Breast Cancer	AG	 Slightly Increased
HNF1B	Prostate Cancer	GG	 Average
PLCE1	Esophageal Cancer	AA	 Average

GENE	CONDITION	GENOTYPE	RISK LEVEL
HERC2/OCA2	Melanoma (fair skin)	AG	 Moderately Increased
SLC45A2	Melanoma (fair skin)	CG	 Moderately Increased
FGFR2	Breast Cancer	AA	 Moderately Increased
2q35	Breast Cancer	GG	 Average
MSMB	Prostate Cancer	TC	 Slightly Increased
SMAD7	Colorectal Cancer	CC	 Average
CHRNA3/CHRNA5	Lung Cancer	GG	 Average
FOXE1	Thyroid Cancer	AG	 Moderately Increased
ABO	Pancreatic Cancer	CT	 Slightly Increased
BNC2	Ovarian Cancer	CC	 Moderately Increased
IRF4	Basal Cell Carcinoma	CC	 Average
MC1R	Skin Cancer / Sun Sensitivity	CC	 Average
IL1B	Gastric Cancer	GA	 Average
KITLG	Melanoma / Testicular Cancer	TT	 Average
TP63	Bladder Cancer	TT	 Average
8q24.21	Bladder Cancer	GG	 Moderately Increased
NKX2-1	Thyroid Cancer	CC	 Average
MAP3K1	Breast Cancer	AC	 Slightly Increased
ESR1	Breast Cancer	GG	 Average
CCND1	Breast Cancer	CC	 Average
TET2	Prostate Cancer	AA	 Average
8q24	Prostate Cancer	GG	 Average
EIF3H	Colorectal Cancer	AA	 Moderately Increased
BMP4	Colorectal Cancer	CC	 Slightly Increased
RHPN2	Colorectal Cancer	CC	 Average
HLA-DRA	Hodgkin Lymphoma	TT	 Average
TERT	Multiple Cancers	CC	 Average
CDH1	Colorectal Cancer	GG	 Slightly Increased
8q24.21	Colorectal Cancer	TT	 Average

Neurological

GENE	CONDITION	GENOTYPE	RISK LEVEL
APOE	Alzheimer's Disease (protective)	CC	 Average
SNCA	Parkinson's Disease	GT	 Slightly Increased
SNCA	Parkinson's Disease	CT	 Slightly Increased
LRRK2	Parkinson's Disease	CG	 Slightly Increased
HLA-DRB5	Alzheimer's Disease	AC	 Slightly Increased
CR1	Alzheimer's Disease	GG	 Average
BDNF	Depression Susceptibility	AG	 Slightly Increased

GENE	CONDITION	GENOTYPE	RISK LEVEL
IL8	Migraine	TA	 Slightly Increased
ADH1B	Alcohol Dependence (protective)	CT	 Low
PICALM	Alzheimer's Disease	TT	 Average
BIN1	Alzheimer's Disease	AA	 Average
PRDM16	Migraine	CC	 Slightly Increased
TRPM8	Migraine	TT	 Moderately Increased
BTBD9	Restless Legs Syndrome	TT	 Significantly Increased
ANK3	Bipolar Disorder	CC	 Average
CACNA1C	Bipolar Disorder	GG	 Average
LINGO1	Essential Tremor	AA	 Moderately Increased
HTR2A	Depression / Schizophrenia	TT	 Average
FKBP5	PTSD / Stress-Related Disorders	TC	 Slightly Increased
CLU	Alzheimer's Disease	TT	 Slightly Increased
MS4A6A	Alzheimer's Disease	AG	 Slightly Increased
CLOCK	Sleep Disorders / Circadian Disruption	AA	 Average
SLC1A2	Essential Tremor	GG	 Average
TREM2	Alzheimer's Disease	CC	 Average
CD2AP	Alzheimer's Disease	GA	 Slightly Increased
MEF2C	Alzheimer's Disease	AG	 Slightly Increased
NME8	Alzheimer's Disease	AA	 Average
AS3MT	Schizophrenia	CC	 Average
MIR137	Schizophrenia	TT	 Slightly Increased
PVRL2/NECTIN2	Alzheimer's Disease	CC	 Average
SORL1	Alzheimer's Disease	TT	 Average
PARK16	Parkinson's Disease	TT	 Average

Autoimmune

GENE	CONDITION	GENOTYPE	RISK LEVEL
HLA-DQ2 . 5	Celiac Disease	CC	 Average
HLA-DRB1	Rheumatoid Arthritis	TT	 Significantly Increased
PTPN22	Type 1 Diabetes	GG	 Average
HLA-DRB1	Multiple Sclerosis	GG	 Average
IL2RA	Multiple Sclerosis	TT	 Average
PTPN2	Crohn's Disease	TT	 Average
IL23R	Crohn's Disease (protective)	GG	 Average
TNFAIP3	Psoriasis	TT	 Average
STAT4	Systemic Lupus Erythematosus	GG	 Average
IL10	Inflammatory Bowel Disease	TC	 Average
CTLA4	Autoimmune Thyroid Disease	AA	 Average

GENE	CONDITION	GENOTYPE	RISK LEVEL
HLA-C	Psoriasis	CC	 Average
HLA-B	Ankylosing Spondylitis	GG	 Average
HLA-DQ8	Celiac Disease	TT	 Average
IL7R	Multiple Sclerosis	CT	 Slightly Increased
TNF	Inflammatory Disease / Sepsis	GG	 Average
SH2B3	Coronary Artery Disease / Autoimmune	TC	 Slightly Increased
TLR4	Inflammatory Bowel Disease / Sepsis	AA	 Average
IL2	Type 1 Diabetes	CA	 Average
NOD2	Crohn's Disease (R702W)	CC	 Average
IL2/IL21	Celiac Disease / Autoimmune (protective)	GG	 Average
TNIP1	Systemic Lupus Erythematosus	AA	 Moderately Increased
HLA-DQA2	Ulcerative Colitis	GA	 Moderately Increased
ATG16L1	Crohn's Disease	GG	 Moderately Increased
MST1	Inflammatory Bowel Disease	AA	 Moderately Increased
BSN	Crohn's Disease	AA	 Slightly Increased
ORMDL3/GSDMB	Crohn's Disease	GG	 Average
2p15	Ankylosing Spondylitis	AA	 Moderately Increased
ERAP1	Ankylosing Spondylitis	AA	 Average
IL23R	Ankylosing Spondylitis	AG	 Slightly Increased
PHOX2B	Crohn's Disease	TT	 Average

Ophthalmology

GENE	CONDITION	GENOTYPE	RISK LEVEL
CDKN2B	Glaucoma	TT	 Moderately Increased
C3	Age-Related Macular Degeneration	GC	 Moderately Increased
GJD2	Myopia (High)	AA	 Slightly Increased
SIX6	Primary Open-Angle Glaucoma	CT	 Moderately Increased
CFI	Age-Related Macular Degeneration	CT	 Slightly Increased
CFI	Age-Related Macular Degeneration	GT	 Slightly Increased
VEGFA	Diabetic Retinopathy	GC	 Moderately Increased

Musculoskeletal

GENE	CONDITION	GENOTYPE	RISK LEVEL
GC	Osteoporosis (via Vitamin D)	TT	 Average
LRP5	Osteoporosis	CC	 Average
TNFSF11	Osteoporosis	CT	 Slightly Increased
TNFRSF11B	Osteoporosis	GG	 Moderately Increased
GDF5	Osteoarthritis	AG	 Average
ESR1	Osteoporosis	CC	 Average

GENE	CONDITION	GENOTYPE	RISK LEVEL
SOST	Osteoporosis	TT	 Average
FAM210A	Osteoporosis / Fracture Risk	TT	 Average
FUBP3	Osteoporosis	CC	 Average
TNFRSF11B	Osteoporosis	TT	 Slightly Increased
LPA	Aortic Valve Calcification	AA	 Average
MIR196A2	Osteoarthritis	CC	 Moderately Increased

Respiratory

GENE	CONDITION	GENOTYPE	RISK LEVEL
GSDMB	Asthma	GG	 Moderately Increased
ORMDL3	Asthma	TT	 Significantly Increased
TSLP	Asthma / Allergic Disease	CC	 Average
ADRB2	Exercise-Induced Bronchodilation / Asthma	AG	 Slightly Increased
LMOD1	Obstructive Sleep Apnea	AA	 Average
HHIP	COPD	CT	 Slightly Increased
HHIP	COPD	CT	 Slightly Increased
AGER	COPD / Lung Function	CC	 Average
ADRB2	Asthma / Obesity	CC	 Average

Renal

GENE	CONDITION	GENOTYPE	RISK LEVEL
CLDN14	Kidney Stones	TT	 Moderately Increased
UMOD	Kidney Stones / CKD	AA	 Average
APOL1	Chronic Kidney Disease	AA	 Average
UMOD	Chronic Kidney Disease	GG	 Moderately Increased
STC1	Chronic Kidney Disease	AT	 Slightly Increased
ALPL	Kidney Stones	GG	 Average
VEGFA	Diabetic Nephropathy	AA	 Average

Longevity

GENE	CONDITION	GENOTYPE	RISK LEVEL
CETP	Cardiovascular Longevity	AG	 Slightly Increased
IL6	Inflammaging / Longevity	CG	 Slightly Increased
PON1	Oxidative Stress / Longevity	CT	 Average
KLOTHO	Cognitive Longevity	TT	 Average
SOD2	Oxidative Stress Management	AG	 Slightly Increased
TERC	Telomere Length	CC	 Average
OBFC1	Telomere Length	AA	 Slightly Increased
NAF1	Telomere Length	GG	 Average

GENE	CONDITION	GENOTYPE	RISK LEVEL
RTEL1	Telomere Length	AA	 Average

Dermatological

GENE	CONDITION	GENOTYPE	RISK LEVEL
WNT10A	Hair Thickness / Ectodermal Features	CC	 Average

Pharmacogenomics

31 variants analyzed • 12 actionable

GENE	ALLELE	GENOTYPE	STATUS	AFFECTED DRUGS
CYP2D6	*4	CT	Normal Function	Codeine, Tramadol, Tamoxifen, SSRIs
CYP2D6	*10	GA	Normal Function	Codeine, Tamoxifen
CYP2C19	*3	GG	Normal Function	Clopidogrel
CYP2C19	*17	CC	Normal Function	Clopidogrel, SSRIs
CYP2C9	*2	CC	Intermediate/Poor Metabolizer	Warfarin, Phenytoin, NSAIDs
CYP3A5	*3	CC	Normal Function	Tacrolimus, Cyclosporine
SLC01B1	*5	TT	Normal Function	Simvastatin, Atorvastatin
DPYD	D949V	TT	Intermediate/Poor Metabolizer	5-Fluorouracil, Capecitabine
HLA-B*5701	proxy	TT	Hypersensitivity Risk	Abacavir
HLA-B*1502	proxy	CG	Normal	Carbamazepine, Oxcarbazepine, Phenytoin
TPMT	*2	CC	Normal Function	Azathioprine, 6-Mercaptopurine
TPMT	*3C	TT	Poor Metabolizer	Azathioprine, 6-Mercaptopurine
TPMT	*3B	CC	Normal Function	Azathioprine, 6-Mercaptopurine
CYP1A2	*1F	AA	Ultrarapid Metabolizer	Caffeine, Theophylline, Clozapine
NAT2	*5	TC	Intermediate Metabolizer	Isoniazid, Hydralazine, Sulfasalazine
IFNL3/IL28B	CC	CC	better response	Peginterferon, Ribavirin
OPRM1	A118G	GA	Intermediate Metabolizer	Morphine, Fentanyl, Codeine
CYP2B6	*6	TT	Intermediate/Poor Metabolizer	Efavirenz, Bupropion, Methadone
CYP2C19	*3	TT	Normal Function	Escitalopram, Sertraline, Citalopram, Amitriptyline
CYP2C19	*17	CC	Normal Function	Sertraline, Escitalopram, Fluoxetine
CYP2D6	*10-like	CG	Normal Function	Adderall (amphetamine), Vyvanse (lisdexamfetamine), Ritalin (methylphenidate - limited), Atomoxetine
CYP2D6	*2xN	CC	Normal Function	Adderall (amphetamine), Vyvanse, Atomoxetine
SLC01B1	*15	TT	Decreased Function	Rosuvastatin, Pravastatin, Simvastatin, Atorvastatin
ABCG2	Q141K	AA	Normal Function	Rosuvastatin, Atorvastatin, Fluvastatin
ABCG2	V12M	CC	Normal Function	Rosuvastatin
CYP3A4	*1B	TT	Normal Function	Atorvastatin, Simvastatin, Lovastatin
CYP3A4	*22	TT	Intermediate/Poor Metabolizer	Simvastatin, Atorvastatin, Lovastatin

GENE	ALLELE	GENOTYPE	STATUS	AFFECTED DRUGS
OPRM1	G-172T	GG	Normal Function	Morphine, Oxycodone
CYP3A5	*7	CC	Normal Function	Tacrolimus, Cyclosporine
CYP1A2	*1C	CC	Intermediate/Poor Metabolizer	Caffeine, Theophylline, Clozapine
CYP1A2	*1D	TT	Normal Function	Caffeine, Theophylline

Carrier Status

3 conditions screened • 0 carrier status detected

GENE	CONDITION	GENOTYPE	STATUS
HFE	Hereditary Hemochromatosis	CC	 Not Carrier
ATP7B	Wilson Disease	GG	 Not Carrier
PAH	Phenylketonuria (PKU)	GG	 Not Carrier

🌟 Traits

239 traits detected

🥛 Lactose Tolerance

Likely lactose intolerant (~85% chance) — you lack the lactase persistence allele
MCM6/LCT (GG)

☕ Caffeine Metabolism

Fast caffeine metabolizer (~45% of people) — caffeine clears quickly, safe to drink coffee later in the day
CYP1A2 (AA)

🍷 Alcohol Flush Reaction

Normal ALDH2 function — no alcohol flush reaction. Most common globally
ALDH2 (GG)

🥬 Bitter Taste Perception

Non-taster — bitter foods like broccoli taste mild to you
TAS2R38 (CC)

👂 Earwax Type

Wet earwax type — the ancestral human state, common in European and African populations (~95%). Typical body odor levels
ABCC11 (CC)

👁️ Eye Color

Green, hazel, or light brown eyes most likely (~50% green/hazel, ~37% brown, ~13% blue)
HERC2 (AG)

🎮 Reward Sensitivity

Typical dopamine receptor density
ANKK1/DRD2 (GG)

🦸 Stress Response (Warrior/Worrier)

Balanced — mix of warrior and worrier traits
COMT (AG)

💡 Emotional Resilience

May be more emotionally sensitive and reactive
TPH2 (TT)

💊 Pain Sensitivity

May need slightly more pain medication for same effect
OPRM1 (GA)

🌙 Sleep Chronotype

Night owl tendency — your circadian clock runs a bit late
CLOCK (TT)

📏 Height

Slightly shorter on average from this variant
HMGA2 (TT)

🌟 Freckling

Moderate freckling tendency
OCA2 (GA)

👨 Male Pattern Baldness

Moderate genetic risk for hair loss
EDA2R (TC)

🍞 Carbohydrate Metabolism

Less efficient
TCF7L2 (TC)

🧠 Memory & Learning

Normal BDNF secretion — exercise will still boost it further!
BDNF (CC)

💪 Pain Tolerance

Your genotype is TC — interpretation pending for this variant.
COMT (TC)

😱 Startle Response

Typical startle response
ADRA2A (CC)

🥗 Cholesterol Response to Diet

May respond particularly well to dietary improvements for cholesterol
PCSK9 (TT)

🥑 Dietary Fat Response

Typical (ε3/ε4)
APOE (CC)

 Asparagus Smell Detection

You can smell asparagus metabolites in urine
OR2M7 (GG)

 Photic Sneeze Reflex (ACHOO)

You may occasionally sneeze from bright light
2q22 (CT)

 Nicotine Dependence Risk

Lower genetic risk for nicotine dependence
CHRNA5 (GG)

 Deep Sleep Quality

Typical sleep depth
ADA (CC)

 Fair Skin / Freckling

Typical pigmentation from this MC1R variant
MC1R (CC)

 Pain Perception

Altered pain perception — may experience pain differently
SCN9A (GG)

 Vitamin B12 Levels

Non-secretor — associated with lower B12 absorption and lower susceptibility to many (not all) norovirus strains
FUT2 (GG)

 Omega-3 Conversion Efficiency

Moderate omega-3 conversion — supplement with fish oil recommended
FADS1 (TC)

 Folate Metabolism (A1298C)

Normal MTHFR A1298C — typical folate metabolism
MTHFR (TT)

 Saturated Fat Sensitivity

Typical response to saturated fat
APOA2 (AA)

 Inflammation Response

Moderate inflammatory response
IL6 (CG)

 Red Hair (D294H)

Typical pigmentation
MC1R (GG)

 Omega-3 Metabolism

Moderate omega-3 conversion efficiency
FADS1 (CT)

 Circadian Rhythm / Chronotype

Morning person tendency — you naturally wake up early and feel alert
CLOCK (AA)

 Bitter Taste Sensitivity (PTC/PROP)

Enhanced bitter taste sensitivity — supertaster territory
TAS2R38 (TT)

 Light Hair / Pigmentation

Tendency toward lighter hair and skin pigmentation
MC1R (GG)

 Skin Pigmentation

Lighter skin pigmentation — European-associated SLC24A5 variant
SLC24A5 (AA)

 East Asian Skin Pigmentation

Lighter skin (East Asian variant) — independent from European skin lightening
OCA2 (TT)

 Earlobe Attachment

Attached earlobes
EDAR (CC)

 Hair Thickness

Finer hair
EDAR (AA)

 Sweet Taste Preference

Higher sweet preference
TAS1R2 (CC)

 Coffee Consumption Tendency

Lower drive
CYP1A1 (CC)

 Alcohol Consumption Tendency

Faster metabolism/less drinking
ADH1B (CT)

 Salt Sensitivity

Moderate
AGT (AG)

 Sleepwalking Risk

Typical
ADA region (GG)

 Mathematical Ability

Associated with higher math aptitude
ROB01 (AA)

 Obesity Risk

Higher obesity risk (+3kg average)
FTO (TT)

 Lean Body Mass

Higher lean mass potential
MSTN (TT)

 Ankle Sprain Risk

Moderate
GDF5 (AG)

 Achilles Tendon Injury

More resilient tendons
COL5A1 (CC)

 Resting Heart Rate

Moderate
GJA1 (TC)

 Blood Pressure Response to Exercise

Moderate
EDN1 (GT)

 Migraine Susceptibility

Higher migraine risk
PRDM16 (CC)

 Telomere Length

Longer telomeres (biological youth)
TERC (CC)

 Seasonal Allergy

Higher allergy risk
HLA-DQB1 (TT)

 Myopia / Nearsightedness

Higher myopia risk
GJD2 (AA)

 Ice Cream Headache / Brain Freeze

Less susceptible
TRPM8 (TT)

 Facial Symmetry

More asymmetry
SUPT3H (AA)

 Dandruff Susceptibility

Higher dandruff risk
ZNF750 (AA)

 Nail Strength

Average
SMARCAD1 (TC)

 Gray Hair Onset

Later graying
KITLG (TT)

 Zinc Metabolism

Moderate
SLC30A8 (AG)

 Vitamin C Requirement (SLC23A1 - rs33972313)

Normal absorption
SLC23A1 (CC)

 Gluten Sensitivity Risk

Lower celiac risk
HLA-DQ2.5 (CC)

 Histamine Intolerance

Normal DAO
ABP1/AOC1 (CC)

 Protein Utilization Efficiency

Better protein metabolism and insulin sensitivity
PPARG (CC)

 Appetite Regulation (Leptin Receptor)

Reduced leptin receptor sensitivity (harder to feel full)
LEPR (GG)

 Blood Clotting Speed

Normal clotting
F5 (CC)

 Touch Sensitivity

Moderate
COMT (AG)

 Common Cold Susceptibility

Normal TLR4 signaling
TLR4 (AA)

 Autoimmune Tendency

Typical immune regulation
PTPN22 (GG)

 Dust Mite Allergy

Elevated IgE response to dust mites
IL13 (CT)

 Pet Allergy Tendency

Higher pet allergy risk (especially cats)
ORMDL3 (TT)

 Eczema / Atopic Dermatitis Risk

Lower risk
IL6R (CC)

 Cavity Risk / Dental Caries

Stronger enamel formation
AMELX (CC)

 Wisdom Teeth Development

Variable
PAX9 (AG)

 Tooth Enamel Strength

Stronger enamel mineralization
ENAM (GG)

 Jaw Shape / Malocclusion

More typical jaw development
ABCA4 region (CC)

 Testosterone Levels

Moderate
CYP19A1 (AG)

 Perfectionism Tendency

More flexible/relaxed approach
HTR2A (AA)

 Impulsivity

More deliberate decision-making
DBH (CC)

 Sprint Speed Potential

Higher iron availability fueling fast-twitch muscle
HFE (CC)

 Swimming Endurance

Enhanced mitochondrial biogenesis (better endurance)
PPARGC1A (TC)

 Muscle Recovery Time

Slower recovery
CILP (GG)

 Migraine with Aura Risk

Lower risk
MTDH (CC)

 Vertigo Susceptibility

Altered potassium channel in vestibular neurons (higher vertigo risk)
KCNK18 (TT)

 Dopamine Baseline Levels

Higher dopamine transporter expression (lower synaptic dopamine)
DAT1/SLC6A3 (CC)

 Psoriasis Risk

Typical risk
HLA-C (CC)

 Keloid Scarring Risk

Higher keloid risk
NEDD4 (TT)

 Thyroid Function

Moderate
FOXO1 (AG)

 Insulin Sensitivity

Mildly reduced
TCF7L2 (TC)

 Cortisol Levels (Stress Hormone)

Lower baseline cortisol
CRH (TT)

 Leptin Sensitivity (Satiety)

Moderate
LEP (AG)

 Asthma Risk

Higher childhood-onset asthma risk
GSDMB (GG)

 Lung Capacity

Faster epoxide hydrolase (better detox, preserved lung capacity)
EPHX1 (CC)

 Respiratory Infection Recovery

Slower recovery
IL10 (TC)

 Disc Degeneration Risk

Typical
CHST3 (TC)

 Carpal Tunnel Susceptibility

Metabolic syndrome tendency (higher CTS risk via tissue swelling)
IRS1 (CC)

 Gestational Diabetes Risk

Higher gestational diabetes risk
KCNQ1 (CC)

 Postpartum Depression Risk

Altered estrogen receptor sensitivity (higher PPD risk)
ESR1 (CC)

 Sunlight Mood Boost (SAD Risk)

More sensitive melanopsin (stronger light-mood link)
OPN4 (CC)

 Sauna / Heat Therapy Response

Moderate
ADRB2 (AG)

 Pace of Life (Walking Speed)

Faster habitual walking speed
CASC17 (AA)

 Collagen Breakdown Speed

Higher collagenase activity (faster collagen loss)
MMP1 (CC)

 Oxidative Stress Resistance (SOD2)

Moderate
SOD2 (AG)

 Stuttering Risk

Slightly higher stuttering risk
GNPTAB (TT)

 Bloating Tendency

Higher bloating tendency
IL2RA (TT)

 Restless Leg Syndrome

Moderate
BTBD9 (AG)

 Social Anxiety Tendency

Lower anxiety tendency
NGF (GG)

 Leadership Drive

More comfortable following
RS4950 (AA)

 Night Eating Tendency

Appetite shuts down normally at night
CLOCK (CC)

 Astigmatism Risk

Your genotype is AC — interpretation pending for this variant.
LAMA2 (AC)

 Depth Perception

Enhanced stereoscopic depth perception
OCA2 (CC)

 Itch Sensitivity

More itch-sensitive
FAAH (CC)

 Numbness / Neuropathy Risk

Moderate
GCH1 (CT)

 Voice Deepening Age (males)

Your genotype is TT — interpretation pending for this variant.
AR (TT)

 Breast Tissue Density

Higher breast density
TOX3 (AG)

 Jet Lag Recovery Speed

Faster circadian reset (recovers from jet lag quicker)
RGS16 (CC)

 Déjà Vu Frequency

Rare déjà vu
CHRNA2 (CC)

 Yawning Contagion

Resistant to yawn contagion
MNS1 (CC)

 Sunburn Sensitivity (SLC45A2)

More resistant to burning
SLC45A2 (CG)

 Altitude Sickness Risk

Better tolerance
TNF (GG)

 Chemical Sensitivity

Moderate
COMT (AG)

 Heavy Metal Detox Efficiency

Reduced detox capacity
GSTP1 (AG)

 Pesticide Sensitivity

Typical
SLC01B1 (TT)

 Thermoregulation Efficiency

Efficient temperature regulation
ADRB2 (CC)

 Pollen Allergy Severity

Higher risk of severe hay fever
HLA-DQ (TT)

 Mold Sensitivity

Typical immune response to mold
ATG16L1 (GG)

 Chlorine Sensitivity

Typical metabolism
CYP2C19 (GG)

 Radiation Sensitivity

Moderate
MTHFR (AG)

 Secondhand Smoke Vulnerability

Typical or e4 carrier, more vulnerable
APOE (CC)

 Cold Tolerance

Less efficient, feels cold easily
DI02 (TC)

 Water Retention Tendency

Moderate
GNB3 (AG)

 Sweat Sodium Content

Your genotype is CC — interpretation pending for this variant.
CFTR (CC)

 Altitude Adaptation

Typical
HIF1A (CC)

 Gluten Sensitivity Genetic Risk

Lower risk
HLA-DQ8 (TT)

 Dairy Protein Sensitivity

Lactase non-persistent, may also react to casein
MCM6 (GG)

 Intermittent Fasting Response (TCF7L2)

May have blood sugar instability during fasts
TCF7L2 (TC)

 Fat-Soluble Vitamin Absorption

Typical
APOE (CC)

 Vitamin A Conversion (BCMO1 - rs12934922)

Moderate
BCMO1 (AT)

 Vitamin D Requirement

Typical
VDR (CT)

 Magnesium Utilization

Moderate
COMT (AG)

 Probiotic Response

Different gut bacteria composition (non-secretor), may respond differently to probiotics
FUT2 (GG)

 Vitamin D Binding Protein

Lower vitamin D binding, may have lower blood levels
GC (TT)

 B12 Metabolism Efficiency

Typical
MTRR (AA)

 Iron Deficiency Susceptibility

Lower hepcidin, higher iron absorption
TMPRSS6 (AA)

 Appetite Regulation (FTO)

Increased appetite and reduced satiety
FTO (TT)

 Snacking Tendency

Higher tendency to snack between meals
FTO (TT)

 Triglyceride Response to Carbs

Typical response
GCKR (CC)

 Metabolism Speed (BMR)

Slightly lower BMR, easier weight gain
ADRB3 (GA)

 Depression Susceptibility

Reduced serotonin autoreceptor function, higher depression risk
HTR1A (GG)

 PTSD Susceptibility

Your genotype is CA — interpretation pending for this variant.
FKBP5 (CA)

 Seasonal Affective Disorder Risk

Typical
CLOCK (CC)

 Addiction Susceptibility

Typical receptor density
DRD2 (GG)

 Bipolar Risk Markers

Altered calcium channel function, associated with bipolar traits
CACNA1C (GG)

 Anxiety Disorder Risk

Reduced brain serotonin synthesis, higher anxiety susceptibility
TPH2 (TT)

 Processing Speed

Typical dopamine receptor density
ANKK1 (GG)

 Brain Plasticity (BDNF)

Typical BDNF secretion and neuroplasticity
BDNF (CC)

 Working Memory Capacity (COMT)

Your genotype is AG — interpretation pending for this variant.
COMT (AG)

 Hippocampal Volume

Slightly smaller
TESC (TT)

 Decision Making Speed

Faster dopamine clearance, quicker decisions
COMT (CC)

 Mathematical Ability (ASPM)

Associated with larger brain regions for numerical processing
ASPM (CC)

 Language Learning Aptitude

Enhanced neural connectivity for language
CNTNAP2 (TT)

 Atrial Fibrillation Risk

Typical risk
PITX2 (CC)

 Stroke Risk

Increased stroke risk
NINJ2 (AG)

 Prothrombin Clotting Variant

Typical
F2 (GG)

 Heart Attack Risk (CAD)

1.3x
9p21 (CG)

 HDL Cholesterol Function

Better HDL function
PON1 (CT)

 Lipoprotein(a) Levels

Typical
LPA (AA)

 Blood Clotting Speed (F12)

Typical
F12 (GG)

 Aspirin Responsiveness

Typical response
LPA (TT)

 Blood Pressure Sensitivity to Salt

Typical
AGTR1 (AA)

 Platelet Reactivity

Typical
ITGB3 (TT)

 Heart Rate Variability

Better HRV
GJA4 (TC)

 Homocysteine Levels

Typical
MTHFR/NPPB (AA)

 CRP / Baseline Inflammation

Lower IL-10 production, higher baseline CRP inflammation
IL10 (GG)

 Lip Thickness

Thinner lips
IRF6 (GG)

 Chin Projection

Less projected
EDAR (TT)

 Hair Thickness (Strand Diameter)

Finer strands
EDAR (AA)

 Under-Eye Dark Circles Tendency

Less pronounced
OCA2 (GA)

 Body Fat Distribution (Apple vs Pear)

More central/abdominal fat storage (apple)
FTO (TT)

 Hand Size

Typical
HMG2 (TT)

 DNA Repair Efficiency

Normal folate-mediated DNA repair
MTHFR (TT)

 Immune System Aging (Immunosenescence)

Faster immune aging
IL6 (CG)

 Oxidative Stress Resistance

Moderate
SOD2 (AG)

 Cellular Repair (Glutathione)

Reduced antioxidant defense
GPX1 (AA)

 Cancer Resistance (General)

More efficient p53 tumor suppressor
TP53 (CC)

 Telomere Length Preservation

Longer telomeres on average
TERC (CC)

 Insulin Sensitivity Preservation

Faster decline
TCF7L2 (TC)

 Cardiovascular Aging Rate

Faster cardiovascular aging
APOC1 (AA)

 Joint Preservation with Age

Better cartilage maintenance
LNK1 (CC)

 Wound Healing Speed

Slower healing
TGFB1 (AG)

 Klotho Anti-Aging Factor

KL-VS variant, higher klotho levels, associated with longevity
KL (TT)

 Ability to Wiggle Ears

Your genotype is TA — interpretation pending for this variant.
MYBPC3 (TA)

 Nail Biting Tendency

Less prone
DRD2 (GG)

 Paper Cut Pain Intensity

Heightened pain nerve sensitivity, more intense paper cut pain
SCN9A (GG)

 Brain Freeze / Ice Cream Headache

Less susceptible
TRPM8 (TT)

 Sleep Talking Tendency

Less likely to sleep talk
DRD2 (GG)

 Teeth Grinding (Bruxism)

Lower risk
PRDM16 (CC)

 Startle Response Intensity

Your genotype is AG — interpretation pending for this variant.
COMT (AG)

 Goosebumps Sensitivity

More reactive piloerection (goosebumps from music/cold/emotion)
ADRB2 (CC)

 Ticklishness (OPRM1)

Altered somatosensory processing, may be more or less ticklish
OPRM1 (GA)

 Ability to Hold Breath

Typical
HIF1A (CC)

 Hangover Severity

May have worse hangovers (impaired metabolism of congeners)
MCM6 (GG)

 Estrogen Metabolism

Typical
INSIG2 (CG)

 PCOS Risk

Lower risk
DENND1A (AA)

 Twin Pregnancy Likelihood

Typical
FSHB (GG)

 Free Testosterone (Bioavailable)

Lower SHBG, more free testosterone available
SHBG (CC)

 Bone Loss Rate (Post-Menopause)

Typical
COL1A1 (CC)

 Synesthesia Tendency

Enhanced cross-modal sensory processing, may experience mild synesthesia
OR2M7 (GG)

 Dream Vividness

Typical dream recall
ADA (CC)

 Hypnosis Susceptibility

Less susceptible
DRD2 (GG)

 Diving Reflex Strength

Stronger mammalian diving reflex (heart rate drops more in cold water)
ADRB2 (CC)

 Ambidexterity Potential

More lateralized
BDNF (CC)

 Phantom Phone Vibration

Your genotype is AG — interpretation pending for this variant.
COMT (AG)

 Caffeine Anxiety / Jitters

Moderate caffeine sensitivity — may get jittery from large doses
ADORA2A (CT)

 Caffeine Sleep Disruption

Moderate sleep disruption from caffeine — cut off by 2 PM
ADORA2A (TC)

 Ability to Smell Flowers (Beta-Ionone)

Higher floral scent sensitivity compared with population average
OR5A1 (GG)

 Taste Bud Density

High taste bud density — supertaster territory with intense flavor perception
CA6 (AA)

 Fear Response Intensity

Typical fear processing and recovery
BDNF (CC)

 Screen / Phone Addiction Tendency

Higher screen addiction susceptibility — your reward circuits crave digital stimulation
CHRNA4 (GG)

 Sun Sensitivity & Freckling

Typical UV response — normal tanning and sun sensitivity
IRF4 (CC)

 Under-Eye Circle Darkness

Moderate
OCA2 (GA)

 Triglyceride Response to Fat

Your genotype is AA — interpretation pending for this variant.
APOA5 (AA)

 B12 Activation Efficiency

Reduced methionine synthase reductase activity, needs more B12
MTRR (AA)

 Vitamin D Transport Efficiency

Efficient vitamin D transport — your binding protein effectively carries vitamin D
GC (GG)

 Fat Burning During Exercise

Enhanced fat burner — your body preferentially uses fat for fuel during exercise
PPARA (GG)

 Beard Patchiness

Moderate coverage
AR/EDA2R (TC)

 Music-Induced Goosebumps (Frisson)

Less likely to experience music frisson
GATA2 (TT)

 Facial Recognition Ability

Better than average face recognition
MCTP2 (CC)

 Cellular Stress Recovery

Pro/Pro, stronger apoptosis response (better cancer defense but faster aging)
TP53 (CC)

 HDL Cholesterol & Heart Longevity

Typical HDL cholesterol levels from this CETP variant
CETP (AA)

 Alcohol Metabolism Speed

Fast alcohol metabolizer — you process ethanol quickly but produce a burst of toxic acetaldehyde
ADH1C (TT)

 Wine Bitterness Perception

Strong tannin bitterness — red wine, dark tea, and grape skins taste intensely bitter to you
TAS2R16 (AA)

 Road Rage / Anger Reactivity

Higher anger reactivity and impulsive aggression
HTR2A (TT)

 Pain Tolerance During Childbirth

Higher pain threshold (may need less epidural)
SCN9A (GG)

 Anesthesia Sensitivity

Moderate
OPRM1 (GA)

 Intermittent Fasting Response

May not respond as well to intermittent fasting
CLOCK (AA)

 ACL Tear Risk

Higher ACL injury risk
COL5A1 (CC)



42 ancestry markers analyzed

Population Composition

European	52%
African	26%
South Asian	20%
East & Southeast Asian	1%
Native American	1%

 **Medical Disclaimer:** GenomeInsight provides genetic information for educational and informational purposes only. This is not a medical diagnostic tool. Results should not be used as a substitute for professional medical advice, diagnosis, or treatment. Genetic risk factors indicate statistical associations, not certainties. Always consult qualified healthcare providers before making medical decisions.

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